

Chapter 1 Checkpoint

(1) Given an account governed by compound interest, what is the relationship between the following parameter? In other words, if I gave you any one of these could you find any other one?

$$i, d, v, i^{(m)}, d^{(m)}, \delta$$

(2) Where does the following expressions move \$5,000 (example could be: forward 3 years or backwards 2 months etc.)

(a) $5000(1+i)^{-4}$

(b) $\frac{5000}{(1-d)^3}$

(c) $5000(1 - \frac{d^{(12)}}{12})^4$

(d) $5000(1 - \frac{i^{(2)}}{2})^{-8}$

(e) $5000v^{-6}$

(3) Would you rather (from an investment percentage) an account governed by...

(a) $i = 7\%$ or $v = 93\%$

(b) $i^{(12)} = 1\%$ or $d = 12\%$

(c) $i = 5\%$ or $\delta = 5\%$

(d) $\frac{i^{(4)}}{4} = 8\%$ or $d^{(4)} = 2\%$

(e) $v = 90\%$ or $i^{(2)} = 5\%$

(4) Given an account governed by simple interest of rate r , find an expression for the following

(a) The discount rate from year 3 to year 6

(b) The interest rate from year 2 to year 4

(c) The interest rate from year 3 to year 5

(d) The force of interest at the start of the 3rd year

(e) The amount of interest earned from year 2 to year 7